

Simulations in Biostatistics: Final Project Work Plan

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Working Title: Evaluating Missing Data Handling Techniques Under Different Missingness Mechanisms

Motivation and Background

In epidemiologic and clinical research, missing data is a common statistical challenge. Whether due to participant dropout, data collection errors, or selective nonresponse, missingness can lead to biased parameter estimates, loss of statistical power, and invalid inferences if not properly addressed. Rubin's (1976) framework defines three mechanisms: Missing completely at random (MCAR) when missingness is independent of all data; Missing at random (MAR) when missingness depends on observed but not unobserved values; and Missing not at random (MNAR) when missingness depends on the missing values themselves. Under MCAR, complete-case analyses yield unbiased estimates, but under MAR or MNAR naive deletion can be seriously biased. Handling missing data properly is therefore necessary to reduce bias, maintain statistical power, and preserve the validity of conclusions.

Planned Objectives of Simulation Workflow

1. Extract summary statistics (mean, SD, and correlation structure) for identified variables from publicly available [UCI Obesity](#) dataset.
2. Introduce missing data under MCAR, MAR, and MNAR mechanisms.
3. Apply and compare imputation techniques:
 - a. Complete Case Analysis
 - b. Mean/Median Imputation
 - c. Multiple Imputation via Chained Equations (MICE)
 - d. Others (TBD based on literature review)
4. Evaluate and compare performance based on:
 - a. Bias in model coefficients
 - b. 95% CI coverage and width
 - c. Log-likelihood, AIC/BIC
 - d. Prediction Accuracy, Calibration
 - e. Others (TBD based on literature review)

Expected Deliverables

- Recorded slide show (20-25 minutes)
- Scientific Report (max 5 pages, R Markdown)

Timeline and Next Steps

Week	Task	Deadline
Jul 14, 2025	Final Project Work Plan	Jul 15, 2025
	Introduction (Literature Review & Motivation)	Jul 17, 2025
	Methods (Data Generating Process)	Jul 18, 2025
Jul 21, 2025	Run simulation experiments	Jul 21, 2025 - Jul 24, 2025
	Results (Prepare slideshow + report drafts)	Jul 25, 2025 - Jul 28, 2025
Jul 28, 2025	Submit slideshow presentation	Jul 29, 2025
	Peer review assessments of presentations	Jul 31, 2025
	Submit final report	Aug 1, 2025